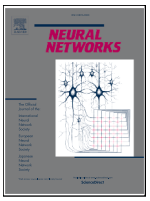
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Full Length Article



Rethinking neural network reparameterized full waveform inversion: A Hessian perspective

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ABSTRACT

By seamlessly integrating deep neural networks (DNNs) into the traditional full waveform inversion (FWI) framework, the neural-network-reparameterized FWI (RFWI) naturally introduces implicit regularization, thereby mitigating key challenges in practical applications—such as sensitivity to initial models and susceptibility to local minima. While a variety of advanced RFWI methods have been proposed to enhance inversion performance, profound insights remain lacking in two key questions: *(1) How does the DNN improve the robustness of RFWI*, *(2) How to further improve the performance of RFWI.* In response to the first question, the *Hessian spectrum* is introduced as an analytical tool to reveal the difference in optimization characteristics between RFWI and traditional FWI. Our findings demonstrate that the superior robustness arises from the persistent constraints imposed by DNN on the inversion process, guiding a long-lasting tendency towards the eigenvectors associated with large eigenvalues, which correspond to stable directions dominated by low-frequency information. For the second question, we propose two improvement techniques based on the inherent optimization characteristics of RFWI, targeting *convergence* or *robustness*, both of which have the potential to generalize to other RFWI implementations. Focusing on *convergence* improvement in ideal scenarios, we propose the Adam + L-BFGS hybrid optimizer, which accelerates high-frequency fitting while avoiding local minima. Regarding *robustness* improvement in challenging scenarios, we propose a Hessian trace-based regularization technique that promotes flatter optimization trajectories, thereby mitigating the overfitting of noise. The experimental results support the effectiveness of our proposed improvement techniques.

1. Introduction

Full waveform inversion (FWI) is a partial differential equation (PDE)-constrained inverse problem in Geophysics (Abedi et al., 2026; Brossier et al., 2009; Tarantola, 1984), which has been successfully extended to various domains, including medical imaging (Guasch et al., 2020), nondestructive testing (Nguyen & Modrak, 2018), and marine sciences (Plessix et al., 2013). The process of FWI is typically formulated as a nonconvex optimization task that iteratively minimizes the misfit between observed data (obtained from field acquisition) and synthetic data (the solution of PDE) to reconstruct high-resolution and accurate subsurface velocity models. By fully exploiting the complete wavefield information, FWI has emerged as a state-of-the-art physics-based approach in seismic imaging.

Despite its remarkable capability and generalization, FWI remains highly non-linear (Virieux & Operto, 2009), with successful convergence strongly dependent on the proximity of the initial model to the true subsurface structure. This nonlinearity often manifests in the form of cycle-skipping, where large discrepancies between the initial and true models can lead to convergence toward incorrect local minima. Furthermore, the limited bandwidth and aperture of seismic data, along with noise contamination, further exacerbate the non-uniqueness and ill-posedness of the inversion (Van den Ende et al., 2021; Iqbal, 2022). To alleviate these limitations, various approaches have been proposed, including multiscale inversion (Bunks et al., 1995; Sirgue & Pratt, 2004), regularization techniques (Asnaashari et al., 2013; Esser et al., 2018), and other approaches to improve convergence stability and solution robustness.

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Recently, deep learning (DL) methods have demonstrated considerable potential in solving inverse problems, owing to their powerful feature representation and hierarchical abstraction capabilities (Dong et al., 2014; Jin et al., 2017; LeCun et al., 2015). These advances offer a promising avenue for addressing the limitations of traditional inversion framework in practical settings. One prominent research direction involves training deep neural networks (DNNs) on large-scale datasets consisting of paired seismic data and corresponding velocity models, thereby enabling direct data-to-model mapping (Araya-Polo et al., 2018; Li et al., 2019; Yang & Ma, 2019). However, such purely data-driven approaches are inherently constrained by the availability of high-quality labeled data, particularly in field applications where ground-truth velocity models are scarce or inaccessible.

To alleviate reliance on labeled data sets and improve the physical consistency of inversion results, researchers have proposed semi-supervised and unsupervised learning approaches that incorporate physical laws into the training process (Jin et al., 2021; Liu et al., 2023). In these approaches, the predicted velocity model is passed through a forward modeling operator to generate synthetic data, and the discrepancy between the synthetic and observed data (i.e., the data misfit) is used to update the network parameters. Despite differences in supervision, these approaches fall within the broader class of learning-based frameworks, all aiming to learn an end-to-end mapping from data to model space.

However, a key limitation persists: Models trained on synthetic datasets often generalize poorly to complex field data due to discrepancies between simulated and real-world seismic conditions. This domain gap significantly hinders the practical applicability of learning-based inversion in real seismic scenarios.

To leverage the generalizability of physics-based methods and the expressive power of DL, researchers have proposed the neural network-reparameterized FWI (RFWI) framework, wherein the velocity model is parameterized by a DNN, and optimization is performed in the network weight space rather than directly on model parameters (He & Wang, 2021; Wu & McMechan, 2019; Zhu et al., 2022). The DNN architecture serves as a regularizer, constraining the solution space and alleviating entrapment in local minima. Specifically, the optimization behavior of DNNs has been demonstrated to inherently follow the frequency principle (or spectral bias) (Rahaman et al., 2019), capturing low-frequency components before progressively refining high-frequency details to enhance representational capability. Such hybrid framework incorporates DNNs into the traditional inversion workflow, sacrificing inference speed but enhancing physical consistency and interpretability via task-specific iterative optimization.

Based on the functional role of neural networks during the inversion process, the RFWI framework can be broadly classified into two paradigms: deep regularization-based RFWI (DR-RFWI) and implicit neural representation-based RFWI (INR-RFWI). The DR-RFWI paradigm is inspired by the concept of deep image prior (DIP), which demonstrates that an over-parameterized DNN (i.e., the number of network parameters far exceeds the output dimension) can inherently capture a great amount of image statistics without any training process (Shu & Entezari, 2024; Ulyanov et al., 2018). Regarding the INR-RFWI paradigm, the velocity model is parameterized as a continuous function, facilitating implicit modeling of the subsurface via a coordinate-based neural network (Sitzmann et al., 2020; Sun et al., 2023). Building upon its remarkable superiority over traditional FWI, RFWI has attracted extensive research attention, inspiring the emergence of various variants under both paradigms.

For DR-RFWI, design choices typically involve three core elements: network input, network architecture, and initialization strategy. Wu and McMechan (2019) and He and Wang (2021) reparameterized the velocity model using convolutional decoders with uniform tensor inputs, highlighting the importance of prior information encoded via pretraining. In contrast, Zhu et al. (2022) demonstrated that untrained networks with random tensor inputs can achieve superior noise robustness. Beyond tensor input configurations, shot gathers have also been explored as direct inputs to the reparameterization architecture. Dhara and Sen (2022) employed an encoder–decoder CNN that directly processed shot gathers. Muller et al. (2023) adopted a U-Net architecture with pretraining based on velocity trends to accelerate convergence. Furthermore, Jiang et al. (2023) utilized a more sophisticated design, SeisInvNet (Li et al., 2019), which also takes shot gathers as input and leverages pretraining to enhance inversion stability.

For INR-RFWI, configuration variants are primarily centered on activation function selection, which plays a critical role in mitigating high-frequency fitting deficiency induced by the frequency principle of DNN. Sun et al. (2023) introduced a sine-activated multilayer perceptron (MLP) to improve the reconstruction of high-frequency components. Zhang et al. (2023) extended the method to elastic inversion scenarios, and emphasized the significant influence of network weight initialization on inversion quality. In light of the frequency-fitting behavior inherent to DNNs, Du et al. (2024) proposed an MLP with the GELU activation function, demonstrating enhanced robustness across diverse conditions compared to the sine-based implementation in Sun et al.'s original work.

Previous studies have achieved substantial progress in the configuration of RFWI methods, demonstrating superior inversion performance compared to traditional FWI under challenging scenarios. Consequently, two intriguing questions naturally emerge that deserve further investigation:

- **How does the DNN improve the robustness of RFWI?**
  A widely accepted statement is that the robustness can be attributed to the implicit regularization (or frequency principle) induced by the network architecture. However, more profound insights into such robustness still remain lacking.
- **How to further improve the performance of RFWI?**
  Some improvement techniques introduced in previous works are often tailored to particular configuration and thus difficult to generalize to other RFWI implementations. Developing improvement techniques that can generalize across different RFWI configurations is of significant importance.

This work focuses on further investigating these unresolved questions from the perspective of RFWI's intrinsic optimization characteristic. Given the lack of consensus regarding the influence of configuration components on inversion performance in the RFWI framework, we first conduct comparative experiments to evaluate the optimal configuration, which serves as the reference setting for subsequent studies. In response to the first question, we introduce the *Hessian spectrum* as a crucial analytical tool to reveal the inherent optimization characteristics of RFWI. Building on our deeper understanding of RFWI in a Hessian perspective, two generalized techniques are proposed to further improve its performance in different application scenarios.

Our contributions can be summarized as follows:

- **Evaluation of Optimal Configuration by Comparative Experiments:** Through comprehensive comparisons of RFWI implementations with varying configurations, we determine the optimal configuration and analyze the impact of the configuration components on inversion performance. Our results and analysis indicate that RFWI performance is largely insensitive to the choice of network input and initialization strategy.
- **Revelation of Network-induced Robustness by Hessian Spectrum:** We employ the *Hessian spectrum* to compare the optimization characteristics between traditional FWI and RFWI. Experimental results show that the constraints imposed by DNN lead the optimization of RFWI primarily along eigenvectors associated with large eigenvalues, corresponding to stable fitting process of low-frequency information.
- **Proposal of Generalizable Improvement Techniques:** For further ***convergence*** improvement in ideal acquisition scenario, we propose

employing the hybrid optimizer of 'Adam followed by L-BFGS' to facilitate the fitting of high-frequency information. To further enhance ***robustness*** in challenging scenarios, we empirically show that optimization process with reduced Hessian trace is more insensitive to high-frequency information, motivating a Hessian-based regularization technique.

2. Background and motivation

In this section, we elaborate on the first- and second-order derivative information of the loss functions (i.e., Gradient and Hessian) in traditional FWI and RFWI. Then, we present the motivation for employing the *Hessian spectrum* as the analytical tool for investigating the two central questions. Both the gradient and the Hessian are described relative to the loss landscape (also called the 'loss surface'), which is defined as the geometric structure of the loss function within the parameter space, illustrating how the loss varies with changes in the parameters (Li et al., 2018).

To facilitate understanding, we first clarify some key concepts discussed in this study. First, **optimization dynamics** denote algorithm's statistical behavior at different parameter states in the loss landscape, including the (1) optimization tendency relative to eigenvectors and (2) the sharpness (or curvature) of the optimization trajectory, both characterized by the *Hessian spectrum*. Second, we emphasize the peculiarities of the *Hessian spectrum* in the context of inversion problems. The **physical interpretation of the Hessian spectrum** specifically refers to the frequency properties of the eigenvectors associated with different eigenvalue magnitudes. Finally, **optimization characteristic** is described by combining the optimization dynamics characterized by *Hessian spectrum* with its physical interpretation, denoting the specific inversion pattern and performance of both RFWI and traditional FWI. A more detailed description and implementation will be provided in Section 4.

2.1. First-order gradient of FWI and RFWI

2.1.1. Traditional FWI

In order to obtain an accurate distribution of velocity in subsurface structures, FWI iteratively minimizes the difference between realistic observed data and synthetic data to update the velocity model parameters. In the acoustic approximation, synthetic data are calculated through forward modeling, which simulates the seismic wavefield propagation governed by the following wave equation:

$$\frac{1}{c^2(\mathbf{r})}\frac{\partial^2 u(\mathbf{r},t)}{\partial t^2} = \nabla^2 u(\mathbf{r},t) + s(\mathbf{r},t) \tag{1}$$

where $\mathbf{r} = (x, z)$ denotes the spatial position in two dimensions, t is time, ∇^2 represents the Laplacian operator, $u(\mathbf{r}, t)$ is the pressure wavefield, $c(\mathbf{r})$ is the acoustic velocity, and $s(\mathbf{r}, t)$ represents the source term.

Generally, the loss function of FWI is expressed in the least-squares form as:

$$\mathcal{L}(\mathbf{m}) = \frac{1}{2}\left\|\mathbf{F}(\mathbf{m}) - \mathbf{d}_{\text{obs}}\right\|_2^2 \tag{2}$$

where $\mathbf{m}$ denotes the model parameters to be updated, $\|\cdot\|_2$ is the L_2 norm, $\mathbf{d_{obs}}$ is the observed data, and $\mathbf{F}$ denotes the forward modeling operator that simulates the physical process of wavefield propagation [i.e., $\mathbf{F}(\mathbf{m})$ denotes the synthetic data $\mathbf{d}_{\text{syn}}$].

In this context, the iterative process of minimizing $\mathcal{L}(\mathbf{m})$ from the initial model ($\mathbf{m}^0$) is usually performed using a gradient-based optimization, which is a local first-order algorithm. The gradient $\nabla\mathcal{L}(\mathbf{m})$ of the model parameters, indicating the direction of the steepest ascent in the loss landscape, can be calculated using the adjoint state method or back propagation, which are mathematically equivalent (Sun et al., 2020; Zhu et al., 2021). The optimization (see Fig. 1(a)) based on the chain rule can be expressed as:

$$\mathbf{m}^{t+1} = \mathbf{m}^t - \eta\nabla\mathcal{L}(\mathbf{m}^t) \tag{3}$$

$$\nabla\mathcal{L}(\mathbf{m}) = \frac{\partial\mathcal{L}(\mathbf{m})}{\partial\mathbf{m}} = \frac{\partial\mathcal{L}(\mathbf{m})}{\partial\mathbf{F}(\mathbf{m})}\frac{\partial\mathbf{F}(\mathbf{m})}{\partial\mathbf{m}} = \mathbf{J}(\mathbf{m})^T\Delta\mathbf{d} \tag{4}$$

where $\mathbf{m}^t$ denotes the model parameters in the t-th time step, η is the step length (or learning rate), the term $\mathbf{J}(\mathbf{m}) = \partial\mathbf{F}(\mathbf{m})/\partial\mathbf{m}$ is the Jacobian matrix of the forward operator $\mathbf{F}$ with respect to the model parameters $\mathbf{m}$, and $\Delta\mathbf{d} = (\mathbf{d}_{\text{syn}} - \mathbf{d}_{\text{obs}})$ represents the data residual.

2.1.2. Neural network reparameterized-FWI

RFWI framework integrates DNN in the skeleton of traditional FWI, and is able to utilize the properties of DNN to mitigate the limitations of traditional FWI, as shown in Fig. 1(b). The velocity model generated by the DNN is mapped to the data domain through forward modeling, and gradients derived from the data misfit are subsequently used to update the network weights. The model parameters reparameterized by the DNN can be formulated as:

$$\mathbf{m} = \mathcal{N}(\mathbf{z}, \boldsymbol{\theta}) \tag{5}$$

where $\mathbf{z}$ denotes the network input, $\boldsymbol{\theta}$ is the learnable network weights, and $\mathcal{N}$ represents the DNN. Thereby, the loss function can be reformulated as:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\left\|\mathbf{F}(\mathcal{N}(\mathbf{z}, \boldsymbol{\theta})) - \mathbf{d}_{\text{obs}}\right\|_2^2 \tag{6}$$

In contrast to the refinement of model parameters in traditional FWI, RFWI transforms the optimization space to the network weight domain. Consequently, the process of minimizing $\mathcal{L}(\boldsymbol{\theta})$ can be performed starting from the initial state (θ^0), which can be randomly initialized or pre-trained:

$$\boldsymbol{\theta}^{t+1} = \boldsymbol{\theta}^t - \eta\nabla\mathcal{L}(\boldsymbol{\theta}^t) \tag{7}$$

$$\begin{aligned}\nabla\mathcal{L}(\boldsymbol{\theta}) = \frac{\partial\mathcal{L}(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}} &= \frac{\partial\mathcal{L}(\boldsymbol{\theta})}{\partial\mathbf{F}(\mathcal{N}(\mathbf{z},\boldsymbol{\theta}))}\frac{\partial\mathbf{F}(\mathcal{N}(\mathbf{z},\boldsymbol{\theta}))}{\partial\mathcal{N}(\mathbf{z},\boldsymbol{\theta})}\frac{\partial\mathcal{N}(\mathbf{z},\boldsymbol{\theta})}{\partial\theta} \\ &= [\mathbf{J}(\mathbf{m})\mathbf{G}(\theta)]^T\Delta\mathbf{d} = \mathbf{J}(\theta)^T\Delta\mathbf{d}\end{aligned} \tag{8}$$

where $\boldsymbol{\theta}^t$ denotes the network weights in the t-th time step, $\mathbf{G}(\theta)$ is the Jacobian matrix of the velocity model parameters generated by the neural network $\mathcal{N}(\mathbf{z}, \boldsymbol{\theta})$ with respect to the network weights θ, so $\mathbf{J}(\theta)$ represents the Jacobian matrix of the forward operator $\mathbf{F}$ with respect to the parameters θ.

2.2. Second-order Hessian of FWI and RFWI

2.2.1. Traditional FWI

By representing the complete second-order derivative information about loss functions, the Hessian matrix serves as a crucial analytical medium for investigating the local geometric properties of the loss landscape. Mathematically, the Hessian matrix of traditional FWI $\mathbf{H}(\mathbf{m})$ can be formulated as:

$$\mathbf{H}(\mathbf{m}) = \frac{\partial^2\nabla\mathcal{L}(\mathbf{m})}{\partial\mathbf{m}\partial\mathbf{m}^T} = \overbrace{\mathbf{J}(\mathbf{m})^T\mathbf{J}(\mathbf{m})}^{\mathbf{H}_{\text{GN}}} + \overbrace{\Delta\mathbf{d}\cdot\frac{\partial^2\mathbf{F}(\mathbf{m})}{\partial\mathbf{m}\partial\mathbf{m}^T}}^{\mathbf{H}_{\text{NE}}} \tag{9}$$

where $\mathbf{H}_{\text{GN}}$ represents the Gauss-Newton term and $\mathbf{H}_{\text{NE}}$ usually represents the non-linear error term associated with the data residual $\Delta\mathbf{d}$ in the Hessian matrix.

2.2.2. Neural network reparameterized-FWI

In RFWI, the Hessian matrix corresponds to the second-order partial derivatives of the loss function with respect to the neural network parameters. Similarly, the Hessian matrix of RFWI $\mathbf{H}(\theta)$ can be approximately expressed as:

$$\mathbf{H}(\theta) = \frac{\partial^2\nabla\mathcal{L}(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T} = \overbrace{\mathbf{J}(\theta)^T\mathbf{J}(\theta)}^{\mathbf{H}_{\text{GN}}} + \overbrace{\Delta\mathbf{d}\cdot\frac{\partial^2\mathbf{F}(\mathcal{N}(\mathbf{z},\boldsymbol{\theta}))}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}}^{\mathbf{H}_{\text{NE}}} \tag{10}$$

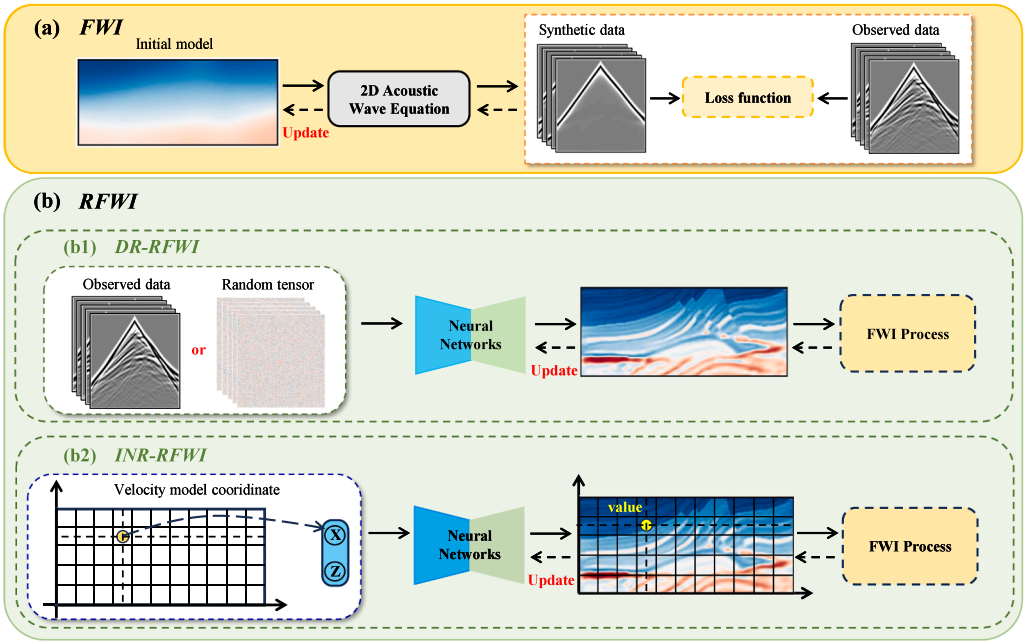


Fig. 1. The frameworks of various inversion methods. (a) Traditional FWI. (b) RFWI.

2.3. Motivation

Our objective is to address two central questions by revealing the intrinsic optimization characteristics of RFWI. Since the gradient only captures a single direction of variation in the parameter space, parameter interactions beyond the gradient direction remain inaccessible. Therefore, relying exclusively on gradient-based analysis can only provide limited insight into optimization characteristics, making it difficult to develop targeted improvement techniques.

Compared to gradients, the Hessian matrix captures the correlations among multiple directions in the loss landscape, offering a broader view to analyze optimization characteristics. Specifically, through the eigendecomposition, the *Hessian spectrum*, comprising all eigenvalues and corresponding eigenvectors, provides a compact representation of the Hessian matrix's principal information:

$$\mathbf{H} = \mathbf{V}\Lambda\mathbf{V}^T \tag{11}$$

where each column of $\mathbf{V}$ represents a specific eigenvector $\mathbf{v}_i$, Λ denotes the diagonal matrix in which the eigenvalues λ_i are arranged along the diagonal entries. Each eigenvector of the *Hessian spectrum* represents a specific parameter variation direction (or pattern) at any stage of the inversion process, and each eigenvalue measures the curvature of the loss landscape along its associated eigenvector. The motivation for employing the *Hessian spectrum* to address our core problems lies in two aspects: (1) optimization dynamics characterization and (2) well-defined physical interpretation.

Optimization dynamics characterization. Employing the *Hessian spectrum* to characterize the optimization dynamics of DNN has become a common practice across numerous problems in DL community. In addition, the optimization dynamics of DNN are often correlated with its performance. A commonly accepted empirical conclusion indicates that DNN with a flatter optimization trajectory, corresponding to regions of reduced sharpness in the loss landscape, tends to yield superior generalization in supervised classification tasks (Keskar et al., 2016; Lee et al., 2023; Sagun et al., 2017). This viewpoint partially explains the success of specific architectural designs (e.g., Batch Normalization layers and residual connections) (Li et al., 2018; Sankar et al., 2021; Yao et al., 2020), and inspires the development of numerous algorithms aimed at enhancing the generalization of DNNs by guiding them toward flatter optimization trajectories (Foret et al., 2020; Wei et al., 2025). This also naturally provides valuable intuition that the sharpness of the DNN optimization trajectory may be related to the inversion characteristic within the RFWI framework.

Well-defined physical interpretation. Based on the peculiarities of the *Hessian spectrum* in geophysical inversion problems, its physical interpretation in this study denotes the frequency properties of the eigenvectors. Specifically, by modifying the parameter state along the eigenvectors in any given stage, the parameter variation patterns represented by eigenvectors can be visualized as the corresponding spatial structural features in the model domain. This provides a basis for analyzing the physical interpretation of *Hessian spectrum*, and further helps us understand the influence of optimization dynamics on the inversion process.

Therefore, the combination of optimization dynamics characterization and well-defined physical interpretation provides a more profound understanding of the RFWI's optimization characteristics from a Hessian perspective. In addition, such insights serve as the foundation for developing generalizable improvement techniques.

3. Preliminary: optimal configuration evaluation

In this section, we evaluate the optimal RFWI configuration through comprehensive comparisons, which serve as a preliminary foundation and establish the reference setting for the subsequent studies.

As noted above, RFWI methods can be classified into two paradigms based on the functional role of neural networks: DR-RFWI and INR-RFWI. Given the differences in paradigm-specific emphases on components of configuration variants, we conduct a two-stage comparative experiment. Specifically, the respective optimal configuration for the two paradigms is first identified. In the second stage, we conduct a comparative evaluation between the optimal configurations of DR-RFWI and INR-RFWI under different scenarios, thereby determining the overall optimal configuration of RFWI.

3.1. Experimental design

3.1.1. Paradigm-specific optimal configuration evaluation

For the DR-RFWI paradigm, our comparison focuses on configuration differences among three critical components:

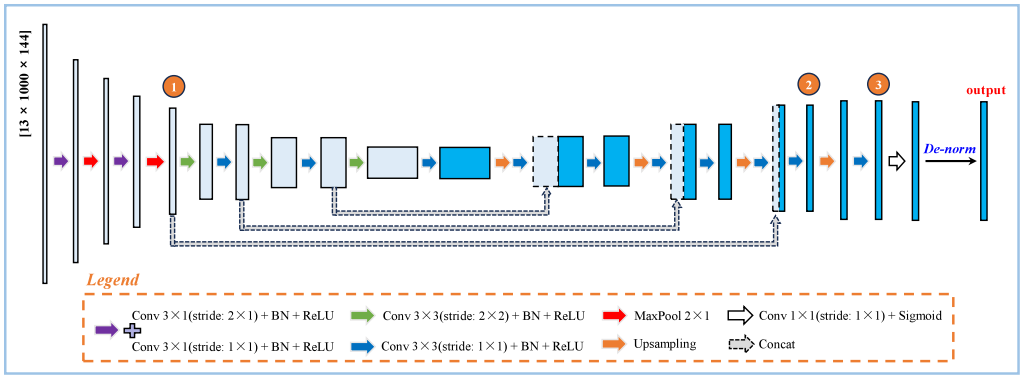


Fig. 2. The architecture operates as the Unet when skip connections are present, and reduces to the AE when these connections are absent.

1. Network Input. For network input, we selected (1) shot gathers (defined as 'Data'), and (2) random tensors sampled from a standard Gaussian distribution (defined as 'Tensor') with dimensional consistency.

2. Network Architecture. We consider and design two common but broadly representative network architectures: (1) CNN with autoencoder architecture (defined as 'AE') and (2) U-Net architecture (defined as 'Unet'), which extends the AE by incorporating skip connections between corresponding feature channels, as shown in Fig. 2. Both architectures maintain approximately 1.7M trainable parameters, substantially exceeding the velocity model dimension.

3. Initialization Strategy. One commonly adopted approach to incorporating prior information is to pretrain the network weights using initial velocity models within a supervised learning scheme. In contrast, some studies have shown that appropriately designed network architectures can eliminate reliance on initial models, achieving superior performance even with randomly initialized network weights (Dhara & Sen, 2022; Zhu et al., 2022). We separately evaluate these two network initialization strategies and formally define them as 'Pretrained' and 'Random'.

For convenience, we define a naming convention of the different DR-RFWI configurations, following the form: *network input—network architecture—initialization strategy*.

Due to the unique operational mechanism of the INR-RFWI paradigm, its configuration variations primarily focus on the selection of activation functions. As such, the experimental design described above is clearly not applicable to INR-RFWI paradigm. Considering the remarkable robustness demonstrated by the inversion configuration of Du et al. (2024), we adopt it as the optimal configuration for the INR-RFWI paradigm. Specifically, their implementation adopts the GELU-activated MLP with the AE network architecture.

3.1.2. Overall optimal configuration evaluation

In the second stage of the experiments, we conduct the qualitative and quantitative evaluation of optimal configurations for DR-RFWI and INR-RFWI. Considering the critical limitations in field data acquisition, we simulate three acquisition scenarios, including (1) noise contamination, (2) low-frequency component deficiency, and (3) limited data. We incorporate traditional FWI inversion results as references to validate the performance enhancement achieved by RFWI.

3.2. Implementation details

The acoustic Marmousi model (Martin et al., 2006) with the grid size of 94 × 288 and the grid spacing of 15 m is selected for all numerical experiments. The ground truth is shown in Fig. 3. The smoothed model serves as the initialization for traditional FWI.

We uniformly deploy 13 sources generated by the Ricker wavelet with a dominant frequency of 8 Hz and 144 receivers at the top of the model. To mitigate artificial boundary reflections, a perfectly matched layer (PML) boundary condition with a thickness of 20 grid points is implemented surrounding the entire model domain. The inversion process takes advantage of the automatic differentiation capability of the PyTorch connected to the Deepwave package (Richardson, 2018) to efficiently compute gradients. The network weights are iteratively updated using the Adam optimizer with a polynomial learning rate scheduling strategy. The initial learning rate is determined through systematic optimization for each experiment to ensure optimal convergence behavior. All numerical experiments are conducted on a single NVIDIA Titan RTX GPU to accelerate computational performance. In addition, we choose the mean squared error (MSE), the structural similarity (SSIM), and the signal-to-noise ratio (SNR) to quantitatively assess the quality of the inversion results.

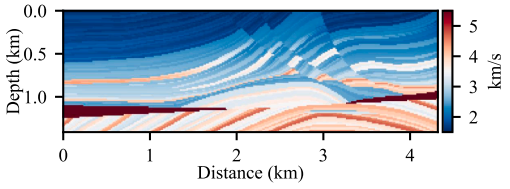


Fig. 3. The acoustic Marmousi model.

In this comparative evaluation, the inversion results obtained by traditional FWI are used as a reference to establish baseline performance under different inversion scenarios. The accuracy of the initial model provided to traditional FWI is one of the key factors in establishing a meaningful reference, as it largely determines whether the inversion process converges to a desirable solution or becomes trapped in a local minimum. To further illustrate this influence, Fig. 4 presents the inversion results of traditional FWI using three commonly used initial models, including a homogeneous model (its velocity value is the average velocity of the ground truth), a linearly increasing velocity model, and a Gaussian-smoothed model. It can be observed that only using the sufficiently accurate smoothed model yields a reasonable inversion result, whereas the other two initializations suffer from severe cycle-skipping effects and fail to recover the main structures.To ensure a meaningful comparison, traditional FWI with a smoothed model initialization is adopted as the baseline for the subsequent experiments.

3.3. Comparison results

3.3.1. Paradigm-specific optimal configuration evaluation

As detailed in Section 3.1, we evaluate the inversion performance of 8 different DR-RFWI configurations, including: (1) *AE—Tensor—Random*, (2) *AE—Data—Random*, (3) *AE—Tensor—Pretrained*, (4) *AE—Data—Pretrained*, 5) *Unet—Tensor—Random*, (6) *Unet—Data—*

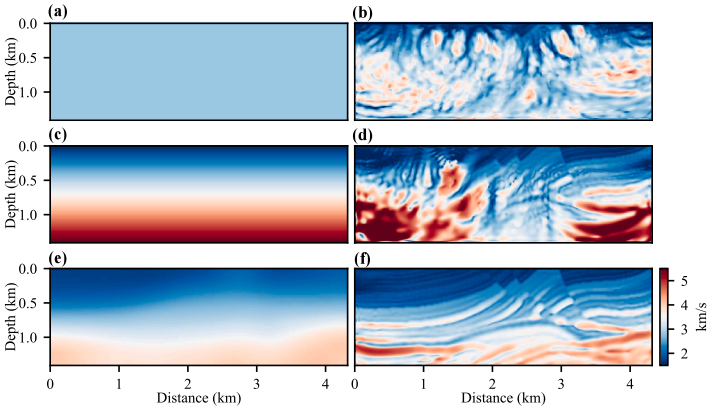


Fig. 4. Different initial models and their corresponding inversion results of traditional FWI. (a) and (b) Homogeneous model. (c) and (d) Linearly increasing velocity model. (e) and (f) Gaussian-smoothed model.

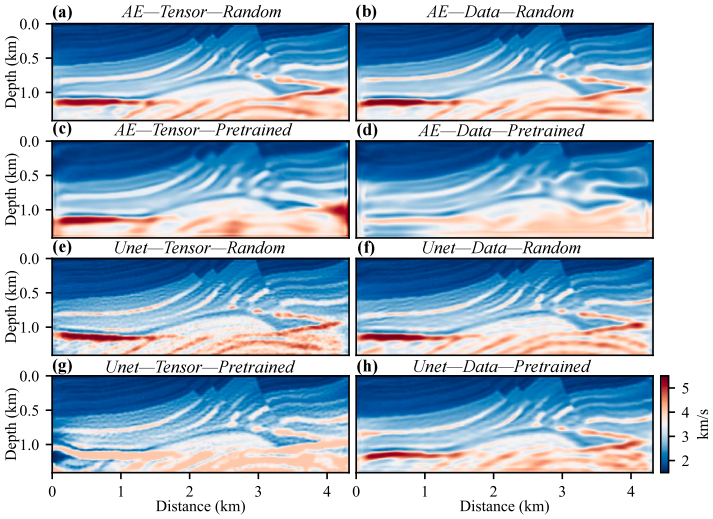


Fig. 5. Comparison of inversion results for different DR-RFWI configurations. (a) *AE—Tensor—Random*. (b) *AE—Data—Random*. (c) *AE—Tensor—Pretrained*. (d) *AE—Data—Pretrained*. (e) *Unet—Tensor—Random*. (f) *Unet—Data—Random*. (g) *Unet—Tensor—Pretrained*. (h) *Unet—Data—Pretrained*.

Random, (7) *Unet—Tensor—Pretrained*, and (8) *Unet—Data—Pretrained*. The inversion results are presented in Fig. 5. Combined with the loss curves shown in Fig. 6, we can observe that all inversion processes achieve the convergent state. In addition, some configurations that initialize the network weights through the pretraining strategy seem to exhibit faster convergence, but yield worse inversion results, such as *AE—Data—Pretrained* and *Unet—Tensor—Pretrained*. The statistical indicators provided in Table 1 support this statement. From an overall perspective, the configurations without pretraining tend to achieve better performance than those relying on the pretraining strategy. The results of the comparison show that the *AE—Tensor—Random* configuration achieves superior inversion performance compared to others, so we establish it as the optimal configuration for the DR-RFWI paradigm. A detailed analysis on the impact of different configuration components on inversion performance is provided in Section 6.

3.3.2. Overall optimal configuration evaluation

To investigate the optimal RFWI configuration, we compare the inversion results of optimal DR-RFWI and INR-RFWI configurations under three challenging scenarios: (1) noise contamination, (2) low-frequency component deficiency, and (3) limited data. We also incorporate traditional FWI results as a reference in the comparisons. Note that we provide the accurate initial model [shown in Fig. 3(b)] for traditional FWI, and randomly initialize the network weights to define the initial state of DR-RFWI and INR-RFWI.

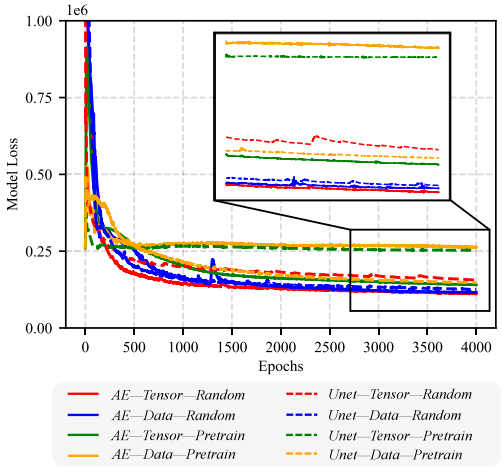


Fig. 6. Loss curves of different DR-RFWI configurations during the inversion process.

Noise contamination. For clarity, the high-quality observed data acquired under the ideal scenario are defined as the reference data. In order to simulate noise contamination in practical applications, we merge Gaussian white noise with mean 0 and standard deviation $\sigma = \sigma_0$, $2\sigma_0$,

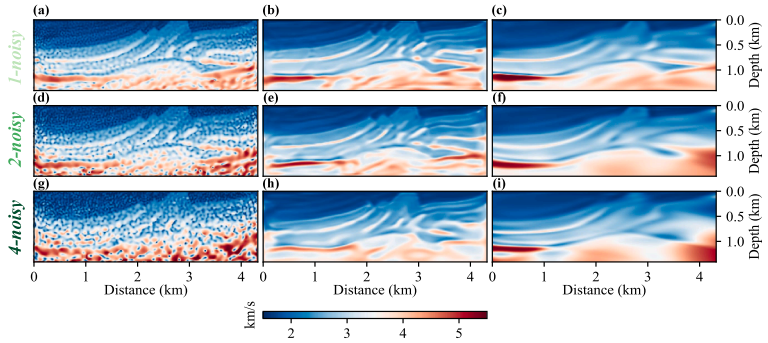


Fig. 7. Comparison of inversion results for different methods in the noise contamination scenarios. From top to bottom, the rows correspond to results obtained using 1-noisy, 2-noisy, and 4-noisy data. From left to right, the columns correspond to the results obtained by traditional FWI, DR-RFWI, and INR-RFWI .

Table 1
Statistics comparison of different DR-RFWI configurations. The ↑ indicates the larger value Achieved, the better performance is, while ↓ indicates the smaller, the better.

	MSE ↓ ($\times 10^6$)	SSIM ↑	SNR ↑
AE—Tensor—Random	**0.1017**	**0.6983**	**19.4992**
AE—Data—Random	0.1098	0.6888	19.1657
AE—Tensor—Pretrained	0.1333	0.5873	18.3301
AE—Data—Pretrained	0.2539	0.5298	15.5365
Unet—Tensor—Random	0.1393	0.6759	18.1456
Unet—Data—Random	0.1138	0.6970	19.0107
Unet—Tensor—Pretrained	0.2519	0.6147	15.5702
Unet—Data—Pretrained	0.1390	0.6383	18.1374

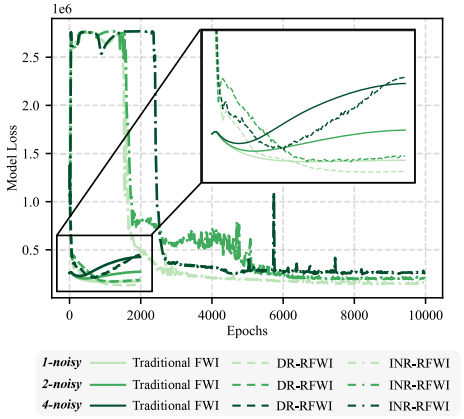


Fig. 8. Loss curves of different inversion methods in the noise contamination scenarios.

Table 2
Statistics comparison of different inversion methods in the noise contamination scenarios.

		MSE ↓ ($\times 10^6$)	SSIM ↑	SNR ↑
1-noisy	Traditional FWI	0.1730	0.5087	17.2106
	DR-RFWI	**0.1334**	**0.6268**	**18.3252**
	INR-RFWI	0.1383	0.5397	18.1675
2-noisy	Traditional FWI	0.2722	0.3861	15.2410
	DR-RFWI	**0.1663**	**0.5870**	**17.3706**
	INR-RFWI	0.1927	0.4959	16.7318
4-noisy	Traditional FWI	0.4259	0.2596	13.2969
	DR-RFWI	**0.2038**	**0.5357**	**16.4897**
	INR-RFWI	0.2368	0.4878	15.8380

Table 3
Statistics comparison of different inversion methods in the low-frequency component deficiency scenarios.

		MSE ↓ ($\times 10^6$)	SSIM ↑	SNR ↑
2 Hz-filtered	Traditional FWI	0.1409	0.6342	18.1015
	DR-RFWI	**0.1182**	**0.6495**	**18.8501**
	INR-RFWI	0.2506	0.5336	15.5928
4 Hz-filtered	Traditional FWI	**0.1682**	0.5965	**17.3298**
	DR-RFWI	0.1751	**0.6131**	17.1475
	INR-RFWI	0.2928	0.5514	14.9189
6 Hz-filtered	Traditional FWI	**0.3182**	**0.4530**	**14.5631**
	DR-RFWI	0.4520	0.4462	13.0389
	INR-RFWI	-	-	-

and $4\sigma_0$ into the reference data, which are defined as 1-noisy, 2-noisy, and 4-noisy data, respectively. The σ_0 represents the standard deviation of the reference data. As shown in Fig. 7, obvious artifacts appear in the inversion results of traditional FWI, and RFWI methods achieve acceptable solutions. The loss curves in Fig. 8 demonstrate that INR-RFWI exhibits significantly longer and more unstable convergence behavior compared to other methods. In addition, DR-RFWI tends to overfit high-frequency noise in the late stage of inversion, which leads to increases in the loss curves. Table 2 indicates that DR-RFWI exhibits optimal noise robustness. Although INR-RFWI seems to be visually less disturbed by noise, DNN is limited by its weak ability to fit high-frequency information in the context of implicit neural representation, resulting in inadequate reconstruction of interface features.

Low-frequency component deficiency. To compare the inversion performance of different methods in the scenarios of low-frequency component deficiency, we process the reference data by removing the low-frequency information below 2 Hz, 4 Hz, and 6 Hz. The corresponding processed data are respectively defined as 2 Hz-filtered, 4 Hz-filtered, and 6 Hz-filtered data. The inversion results shown in Fig. 9 illustrate that RFWI methods without pretraining are more severely affected by the deficiency of low-frequency information, especially for INR-RFWI. In contrast, the DNNs in the DR-RFWI paradigm exhibit superior regularization capability compared with that in the INR-RFWI paradigm. As shown in Fig. 9 (i), the failure of inversion using 6 Hz-filtered data indicates that INR-RFWI is more sensitive to the deficiency of low-frequency components in the observed data. Fig. 10 shows the evolution of model loss, and Table 3 provides quantitative statistics. As can be seen, DR-RFWI is able to achieve reasonable inversion results without any prior information, and its performance remains comparable to that of traditional FWI initialized with a smoothed model.

To more fairly and intuitively compare the robustness of RFWI and traditional FWI in scenarios lacking low-frequency components, we further evaluated traditional FWI with a linearly increasing velocity model as the initial model, thereby removing the advantage provided by a favorable initial model. The qualitative comparison of the results obtained by different methods using 4 Hz-filtered data is shown in Fig. 11. It can be observed that almost no reasonable geological structures can be iden-

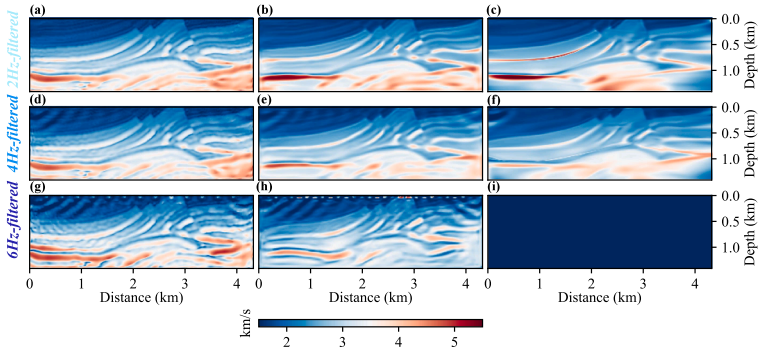


Fig. 9. Comparison of inversion results for different methods in the low-frequency component deficiency scenarios. From top to bottom, the rows correspond to results obtained using 2Hz-filtered, 4Hz-filtered, and 6Hz-filtered data. From left to right, the columns correspond to the results obtained by traditional FWI, DR-RFWI, and INR-RFWI. (i) is a failed inversion result of INR-RFWI using 6 Hz-filtered data.

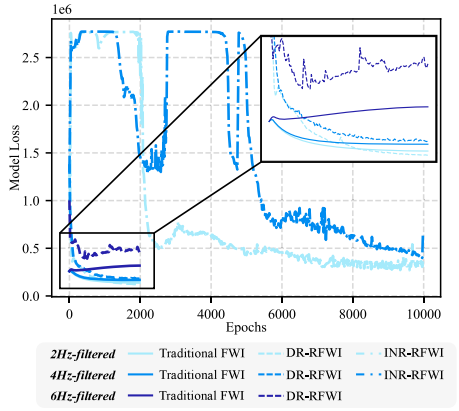


Fig. 10. Loss curves of different inversion methods in the low-frequency component deficiency scenarios.

tified in the inversion result of traditional FWI using a linearly increasing initial model. This indicates that the deficiency of low-frequency component in the observed data further exacerbates the sensitivity of traditional FWI to the initial model. In contrast, RFWI methods can still maintain more stable inversion performance under the same challenging initialization condition. Nevertheless, it should be acknowledged that the increased level of low-frequency component deficiency directly leads to the deviation of the bottom structural features from reality in the DR-RFWI's results and the failure of INR-RFWI. It is foreseeable that the deficiency of low-frequency information will be an unavoidable challenge for the extension of the RFWI framework to practical applications.

Limited data. In order to examine the stability of the inversion process in limited data scenarios for different methods, we reduce the number of sources to 10, 7, and 5, and the corresponding scenarios are defined as 10-shots, 7-shots, and 5-shots data. The inversion results in Fig. 12 show that compared to the model perturbations in the other methods, DR-RFWI consistently exhibits superior stability. As illustrated in Fig. 13, the INR-RFWI requires progressively more iterations to stabilize the inversion process as the constraints derived from the observed data gradually diminish. Table 4 shows an interesting phenomenon: in the 10-shots scenario, INR-RFWI exhibits the reduced MSE and SSIM compared to DR-RFWI. This indicates that INR-RFWI achieves a more accurate velocity distribution relative to ground truth compared to DR-RFWI, but shows inferior performance in interface representation. This phenomenon can be directly attributed to the inherent frequency principle associated with implicit neural representations.

In summary, the results of comparative experiments demonstrate that the *AE—Tensor—Random* configured DR-RFWI has been validated as the optimal RFWI configuration. We select it as the reference setting in subsequent studies and refer to it as the RFWI method hereafter.

Table 4
Statistics comparison of different inversion methods in the limited data scenarios.

		MSE ↓ ($\times 10^6$)	SSIM ↑	SNR ↑
10-shots	Traditional FWI	0.1393	0.6331	18.1479
	DR-RFWI	0.1076	**0.6711**	19.2516
	INR-RFWI	**0.1058**	0.6153	**19.3285**
7-shots	Traditional FWI	0.1518	0.6177	17.7787
	DR-RFWI	**0.1052**	**0.6833**	**19.3525**
	INR-RFWI	0.1328	0.5613	18.3420
5-shots	Traditional FWI	0.1702	0.5862	17.2792
	DR-RFWI	**0.1094**	**0.6671**	**19.1819**
	INR-RFWI	0.1999	0.4988	16.5749

4. How does the DNN improve the robustness of RFWI

In this section, we employ the *Hessian spectrum* to answer the first central question: **How does the DNN improve the robustness of RFWI**. Specifically, for the **optimization dynamics** characterization, we project the parameter update vectors onto the eigenvectors in different parameter states, thereby capturing the optimization directional tendencies relative to each eigenvector. To investigate the **physical interpretation of Hessian spectrum**, we transform the parameter variations represented by the eigenvectors into structural features of the velocity model, enabling a quantitative evaluation of the frequency properties of the eigenvectors. Synthesizing the findings from these two aspects, we reveal the differences in optimization characteristics between RFWI and traditional FWI, which serve as the evidence for answering this question.

4.1. Analytical strategy

4.1.1. Optimization dynamics of RFWI and FWI

In the loss landscape, the local geometry can be effectively represented by eigenvalues and eigenvectors. In the geometric sense, the collection of eigenvectors provides an orthonormal basis corresponding to any parameter state (i.e., the parameters at a given iteration) in the loss landscape. Therefore, each parameter update vector can be decomposed into a weighted linear combination of all eigenvectors, and the corresponding projection factors quantify the degree of alignment between the parameter update and each eigenvector. Taking traditional FWI as an example, the weight associated with each eigenvector can be computed:

$$\Delta \mathbf{m}^t = \mathbf{m}^{t+1} - \mathbf{m}^t = \sum_{i=1}^{M} \alpha_i^t \mathbf{v}_i^t \tag{12}$$

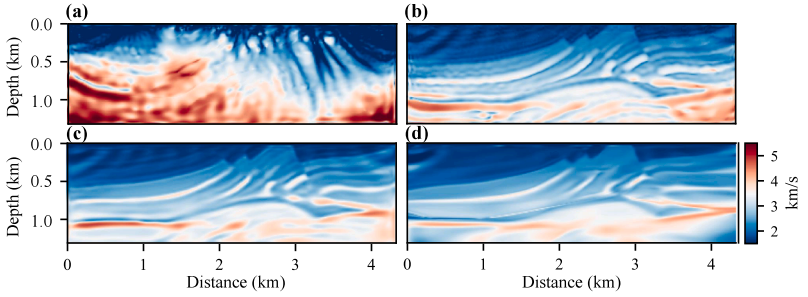


Fig. 11. The inversion results of different methods using 4 Hz-filtered data. (a) Traditional FWI with a linearly increasing initial model. (b) Traditional FWI with a Gaussian-smoothed. (c) DR-RFWI. (d) INR-RFWI.

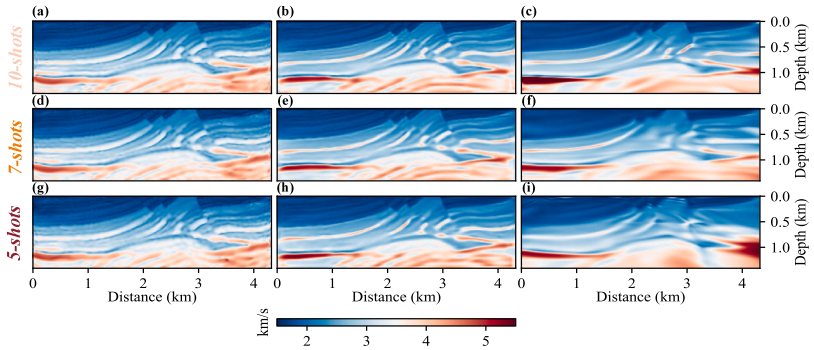


Fig. 12. Comparison of inversion results for different methods in the limited data scenarios. From top to bottom, the rows correspond to inversion results obtained using 10-shots, 7-shots, and 5-shots data. From left to right, the columns correspond to the results obtained by traditional FWI, DR-RFWI, and INR-RFWI.

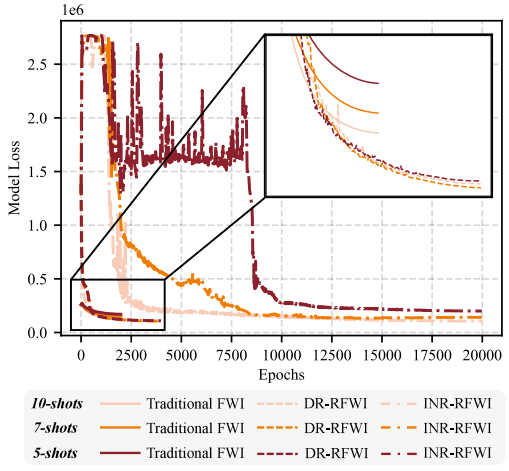


Fig. 13. Loss curves of different inversion methods in the limited data scenarios.

$$\alpha_i^t = (\mathbf{v}_i^t)^T \Delta \mathbf{m}^t \tag{13}$$

where M is the number of eigenvectors, $\Delta \mathbf{m}^t$ represents the parameter update at the t-th iteration, and α_i^t represents the i-th projection factor of the parameter update onto the corresponding eigenvector $\mathbf{v}_i^t$ at the t-th iteration. The characterization is equally applicable to RFWI, as its formal difference from traditional FWI lies solely in the target parameters. The statistical distribution of projection factors associated with eigenvectors describes the optimization directional tendency, indicating that the current update direction exhibits stronger alignment with eigenvectors associated with larger projection magnitudes.

Due to the high dimensionality of the parameters in both traditional FWI and RFWI, explicitly forming and storing the full Hessian matrix to complete set of eigenvectors is infeasible, especially for RFWI. Following the implementation provided in the PyHessian toolkit (Yao et al., 2020), the power iteration method is employed to estimate the eigenvector subset associated with the leading M_s eigenvalues. Eigenvalues are arranged in descending order, with index 1 corresponding to the largest eigenvalue. The detailed computational procedure is presented in Algorithm 1.

4.1.2. Physical interpretation of Hessian spectrum

As mentioned above, the parameter variation patterns represented by eigenvectors can be visualized as the corresponding spatial structural features in the model domain:

$$\mathbf{E}(\mathbf{m})_i^t = (\mathbf{m}^t \pm \alpha_i^t \mathbf{v_i}^t) - \mathbf{m} = \pm \alpha_i^t \mathbf{v}_i^t \tag{14}$$

$$\mathbf{E}(\boldsymbol{\theta})_i^t = \mathcal{N}(\mathbf{z}, \boldsymbol{\theta}^\mathbf{t} \pm \alpha_i^t \mathbf{v}_i^t) - \mathcal{N}(\mathbf{z}, \boldsymbol{\theta}^\mathbf{t}) \tag{15}$$

where $\mathbf{E}(\mathbf{m})_i^t$ and $\mathbf{E}(\boldsymbol{\theta})_i^t$ denote the structural features in the model domain corresponding to the specific eigenvector of traditional FWI and RFWI at the t-th iteration, respectively. This enables us to investigate the frequency properties of different eigenvectors. To this end, we introduce the low-frequency energy ratio (LFER) as an quantitative indicator, which can be expressed as:

$$\text{LFER} = \frac{\sum_{f_i < f_c} [FFT(\mathbf{E_i^t})]^2}{\sum_{f_i < f_{max}} [FFT(\mathbf{E}_i^t)]^2} \tag{16}$$

where FFT denotes the two-dimensional Fourier transform operator, $[FFT(\mathbf{m})]^2$ represents the energy spectrum of the structural features in the model domain projected from the eigenvectors, and f_c is the low-frequency threshold, defined as 0.1 times the Nyquist frequency.

4.2. Experimental results

Fig. 14 presents the Hessian spectrum information of traditional FWI and RFWI. It can be observed that the eigenvalues of both traditional FWI and RFWI are predominantly concentrated around zero in their overall distributions, and the sharpness of the optimization trajectory

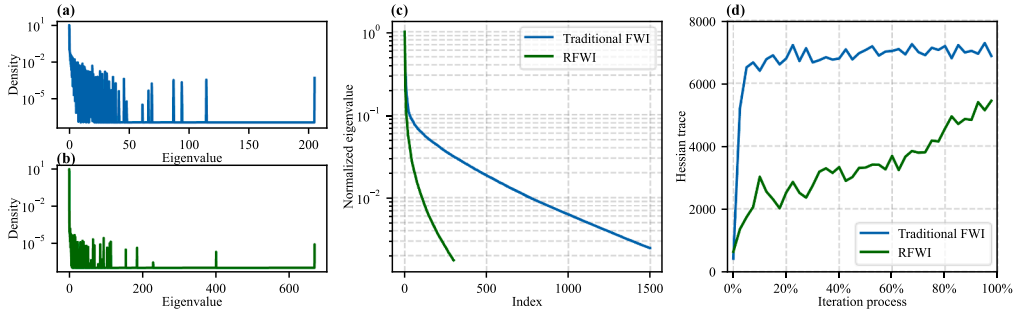


Fig. 14. Comparison of the Hessian spectral information between traditional FWI and RFWI. (a) and (b) Hessian empirical spectral density. (c) Normalized eigenvalue. (d) Hessian trace.

Algorithm 1: Power iteration for estimating leading eigenvalues and eigenvectors

Input : Model parameters $\mathbf{m}$; Loss function $\mathcal{L}$; Convergence tolerance τ; Target number of eigenvectors M_s;
Output: Eigenvalue subset $\Lambda_s = [\lambda_1, \lambda_2, \ldots, \lambda_{M_s}]$;
Eigenvector subset $\mathbf{V}_s = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{M_s}]$;

$\Lambda_s \leftarrow [\,]$, $\mathbf{V}_s \leftarrow [\,]$;
Compute gradient via backpropagation: $\nabla\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\theta}$;
for $i = 1$ **to** M_s **do**
  $\lambda_i \leftarrow$ None;
  Draw a random vector: $\mathbf{v}_i \sim \mathcal{N}(0,1)$; Normalize: $\mathbf{v}_i \leftarrow \mathbf{v}_i/\|\mathbf{v}_i\|_2$;
  for $j = 0$ **to** ∞ **do**
    /* Step 1: Gram-Schmidt Orthogonalization
    for $\mathbf{u} \in \mathbf{V}$ **do**
      $\mathbf{v}_i^{(j)} \leftarrow \mathbf{v}_i^{(j)} - [(\mathbf{v}_i^{(j)})^T\mathbf{u}]\mathbf{u}$;
    Normalize:$\mathbf{v}_i^{(j)} \leftarrow \mathbf{v}_i^{(j)}/\|\mathbf{v}_i^{(j)}\|_2$;
    /* Step 2: Compute Hessian-Vector Product
    Compute: $\nabla\mathcal{L}^T\mathbf{v}_i^{(j)}$;
    Compute: $\mathbf{H}\mathbf{v}_i^{(j)} = \partial(\nabla\mathcal{L}^T\mathbf{v}_i^{(j)})/\partial\mathbf{m}$; // H is Hessian matrix
    /* Step 3: Evaluation Rayleigh Quotient
    $\lambda_i^{(j)} \leftarrow (\mathbf{v}_i^{(j)})^T\mathbf{H}\mathbf{v}_i^{(j)}$;
    /* Step 4: Update and Normalize Trial Vector for Next Iteration
    $\mathbf{v}_i^{(j+1)} \leftarrow \mathbf{H}\mathbf{v}_i^{(j)}/\|\mathbf{H}\mathbf{v}_i^{(j)}\|_2$;
    /* Step 5: Convergence Check
    if $|\lambda_i^{(j)} - \lambda_i^{(j-1)}| < \tau$ **then**
      $\lambda_i \leftarrow \lambda_i^{(j)}$;
      break ; // Eigenvalue converged
  /* Step 6: Store Converged Component
  $\Lambda_s \leftarrow \Lambda_s \cup \{\lambda_i\}$;
  $\mathbf{V} \leftarrow \mathbf{V} \cup \{\mathbf{v}_{i+1}\}$;
return Λ_s, $\mathbf{V}_s$;

shows an overall increasing trend. In comparison, the eigenvalues of RFWI decay more rapidly than those of traditional FWI.

As discussed above, we characterize the optimization dynamics using the subset of eigenvectors corresponding to the leading M_s eigenvalues. In our experiments, the cumulative sum of the leading M_s eigenvalues accounts for approximately 90% of the Hessian trace during the first 20% of the optimization process and more than 95% thereafter. These projection factors are normalized by their maximum absolute value to facilitate clearer visualization, since only their relative magnitudes are required to characterize the optimization tendency.

The optimization tendencies of traditional FWI and RFWI are presented in Figs. 15 and 16, respectively. As shown in Fig. 15, parameter updates in the initial 5% of the optimization process are oriented toward the eigenvectors corresponding to large eigenvalues. As the inversion proceeds, the parameter update behavior rapidly shifts toward the eigenvectors linked to small eigenvalues. This trend emerges at the 20% optimization stage and becomes increasingly pronounced as the inversion continues. In contrast, RFWI exhibits optimization dynamics that are markedly different from those of traditional FWI, as shown in Fig. 16. Although projection factors onto the eigenvectors associated with small eigenvalues gradually increase as the inversion progresses, RFWI updates continue to primarily focus on the eigenvectors corresponding to large eigenvalues before 60% of the optimization process. At the 75% optimization stage, the parameter update shows significant projections along the eigenvectors linked to small eigenvalues, which grow rapidly in subsequent inversion. Nevertheless, before the 95% optimization stage, the projection factors of the parameter updates onto the eigenvectors associated with large eigenvalues remain greater than those onto the eigenvectors associated with small eigenvalues. In summary, parameter updates of traditional FWI tend to follow the eigenvectors associated with small eigenvalues for the majority of the inversion process, while RFWI demonstrates a dominant tendency toward the eigenvectors corresponding to large eigenvalues.

To assess the reliability of the presented optimization tendencies, we constructed a synthetic vector Δm_s^t as the weighted combination of the selected eigenvectors and their corresponding projection factors at t-th iteration:

$$\Delta\mathbf{m}_s^t = \sum_{i=1}^{M_s} \alpha_i^t \mathbf{v}_i^t \tag{17}$$

We then compute the cosine similarity between $\Delta\mathbf{m}_s^t$ and the parameter update vector $\Delta\mathbf{m}^t$ (CS) using the following equation to represent their directional alignment:

$$\mathrm{CS} = \frac{\Delta\mathbf{m}_s^t \cdot \Delta\mathbf{m}^t}{\|\Delta\mathbf{m}_s^t\|_2 \|\Delta\mathbf{m}^t\|_2} \tag{18}$$

As shown in Fig. 17, the CS of RFWI remains high and stable before the 75% optimization stage. This is consistent with the optimization tendency described above, namely that the parameter updates of RFWI are primarily aligned with the eigenvectors associated with large eigenvalues. After the 75% optimization stage, the CS decreases rapidly, indicating that the optimization tendency progressively shifts toward eigenvectors associated with smaller eigenvalues outside the analyzed subset. For traditional FWI, the cosine similarity decreases sharply during the first 20% of the optimization process and remains consistently low throughout the subsequent stages. This observation is also consistent with the evolution of the optimization tendency of traditional FWI illustrated in Fig. 15. Therefore, the evolution of the CS for both RFWI and traditional FWI provides supporting evidence for our description of theire optimization dynamics.

Furthermore, we employ the LFER indicator to quantitatively evaluate the frequency properties of eigenvectors. In order to faithfully evaluate the frequency properties of different eigenvectors, we used the original projection factors rather than the normalized ones as the

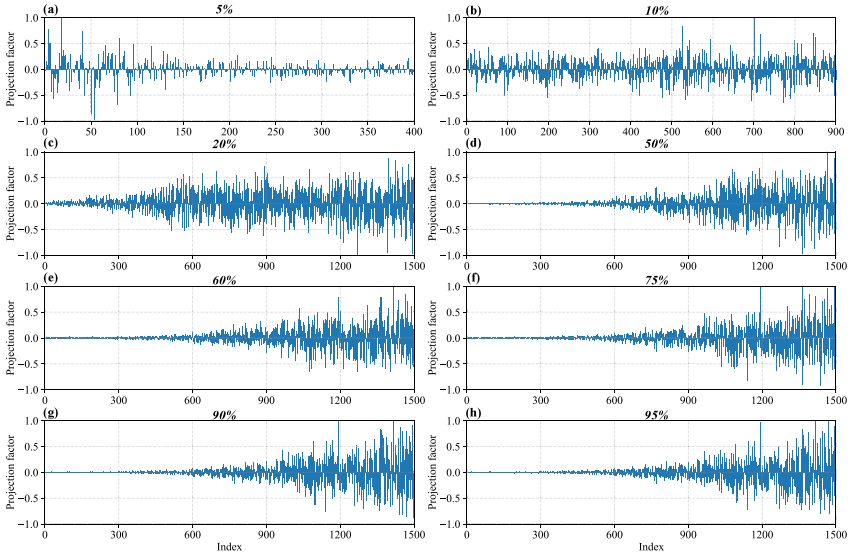


Fig. 15. The optimization directional tendencies of traditional FWI at different optimization stages. The percentage '%' denotes the relative position of the current optimization stage within the overall optimization process. (a) 5%. (b) 10%. (c) 20%. (d) 50%. (e) 60%. (f) 75%. (g) 90%. (h) 95%.

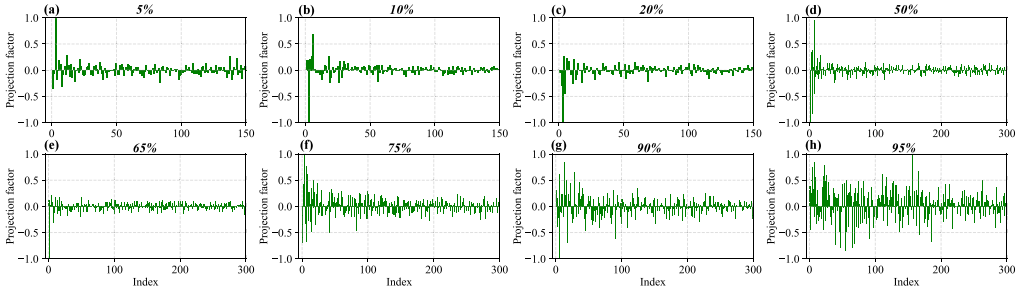


Fig. 16. The optimization directional tendencies of RFWI at different optimization stages. The percentage '%' denotes the relative position of the current optimization stage within the overall optimization process. (a) 5%. (b) 10%. (c) 20%. (d) 50%. (e) 60%. (f) 75%. (g) 90%. (h) 95%.

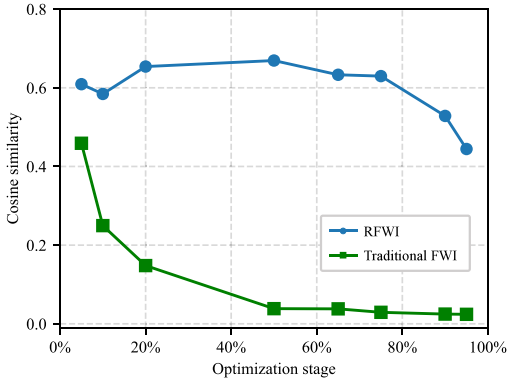


Fig. 17. Evolution of cosine similarity between the synthetic vector $\Delta \mathbf{m}_s^t$ and the parameter update vector $\Delta \mathbf{m}^t$ in traditional FWI and RFWI.

scales of the eigenvectors when visualizing the spatial structural features. Examples of the spatial features represented by the eigenvectors and their corresponding energy spectra are shown in Fig. 18. The frequency properties of eigenvectors indicated by LFER of traditional FWI and RFWI are presented in Figs. 19 and 20, respectively. In an overall perspective across traditional FWI and RFWI, we empirically observe that the spatial features represented by the corresponding eigenvectors become increasingly higher-frequency as the eigenvalues decrease.

By synthesizing the results from the above analyzes, the respective optimization characteristics of traditional FWI and RFWI can be elucidated. Traditional FWI exhibits the optimization pattern focused on fitting high-frequency information from the early stage. In contrast, RFWI exhibits the inversion pattern that maintains a sustained preference for low-frequency information during the majority of the inversion process, thereby preventing premature emphasis on high-frequency components. According to the nature of the inversion problem, the optimization characteristics of RFWI benefit the inversion results in two aspects. First, RFWI can adequately fit the low-frequency components of the observed data during the early stage. Second, RFWI exhibits a gradual incorporation of high-frequency information, which are more strongly affected by noise in noisy scenario. Such optimization characteristics substantially enhance the robustness of RFWI over traditional FWI, thus reducing the risk of becoming trapped in local minima.

Therefore, our findings answer the first key question: the persistent constraint imposed by the DNN limits the optimization tendency of RFWI toward eigenvectors associated with small eigenvalues, thereby suppressing its sensitivity to high-frequency components. This also indicates that RFWI differs significantly from traditional FWI in terms of ***robustness*** and ***convergence***:

- ***Robustness***: RFWI can overcome its dependence on the initial model and achieve acceptable inversion results even under challenging acquisition scenarios.

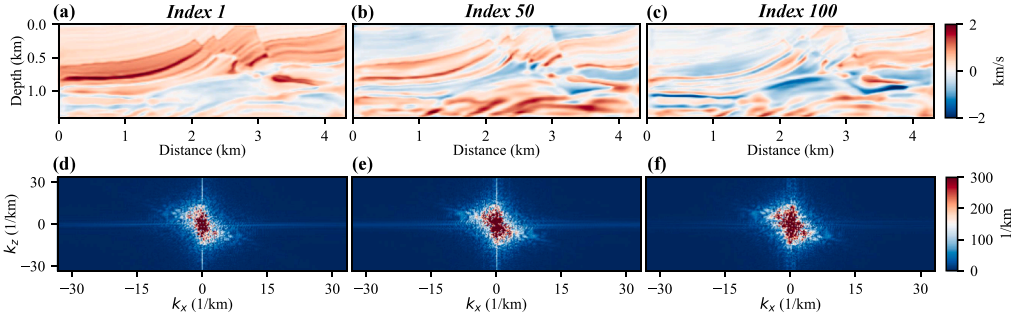


Fig. 18. The spatial features represented by the eigenvectors of RFWI and their corresponding energy spectra. (a) and (d) The results corresponding to the 1-*st* eigenvalue. (b) and (e) The results corresponding to the 50-th eigenvalue. (c) and (f) The results corresponding to the 100-th eigenvalue.

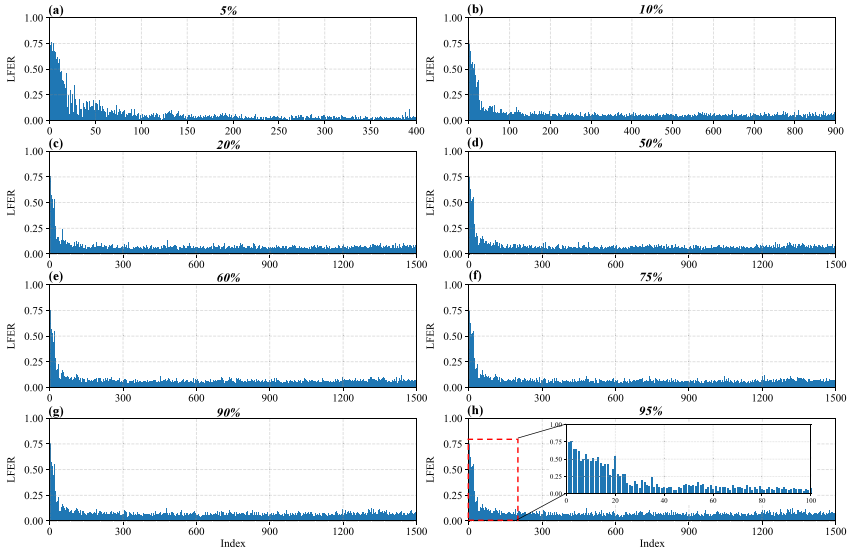


Fig. 19. The frequency properties of FWI eigenvectors at different optimization stages, represented by LFER. The percentage '%' denotes the relative position of the current optimization stage within the overall optimization process. (a) 5%. (b) 10%. (c) 20%. (d) 50%. (e) 60%. (f) 75%. (g) 90%. (h) 95%.

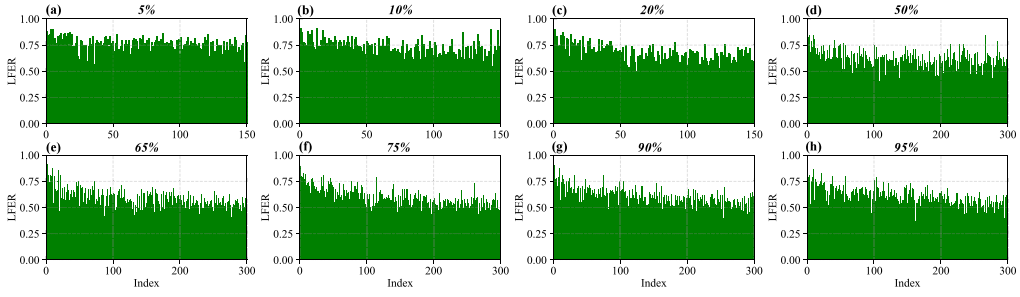


Fig. 20. The frequency properties of RFWI eigenvectors at different optimization stages, represented by LFER. The percentage '%' denotes the relative position of the current optimization stage within the overall optimization process. (a) 5%. (b) 10%. (c) 20%. (d) 50%. (e) 60%. (f) 75%. (g) 90%. (h) 95%.

- ***Convergence***: RFWI shows a slower convergence of high-frequency components compared to traditional FWI.

5. How to further improve the performance of RFWI

Previous studies introduced additional techniques to improve RFWI performance, yet some techniques are only applicable to specific configurations, limiting their generalization to other implementations. For example, while the pretraining strategy has proven effective in some studies (Muller et al., 2023; Wu & McMechan, 2019), it showed the opposite effect in our experiments presented in Section 3.

In this section, we propose targeted techniques to further improve the ***convergence*** and ***robustness*** of RFWI under two different scenarios. It is worth noting that the proposed improvement techniques are grounded in the optimization characteristics of RFWI discussed in Section 4, thus exhibiting greater potential for generalization across different configurations.

5.1. Convergence improvement

In ideal acquisition scenarios with high-quality data, accelerating RFWI ***convergence*** is desired to reduce computational costs. As verified by our findings, RFWI shows insensitivity to eigenvectors associated with small eigenvalues, which limits its convergence rate in fitting high-frequency information. This insight motivates us to improve the RFWI ***convergence*** by promoting the optimization tendencies along eigenvectors associated with small eigenvalues.

Table 5
Statistics comparison of inversion results using different optimizers.

	MSE ↓ ($\times 10^6$)	SSIM↑	SNR↑	Computational Time (min)↓
L-BFGS	0.3548±0.0962	0.5380±0.0431	14.2483±1.2001	63.77
Adam	0.1236±0.0260	**0.6832±0.0044**	19.1420±0.4307	12.00
Adam + L-BFGS	**0.1029±0.0044**	0.6807±0.0095	**19.4528±0.1848**	**10.18**

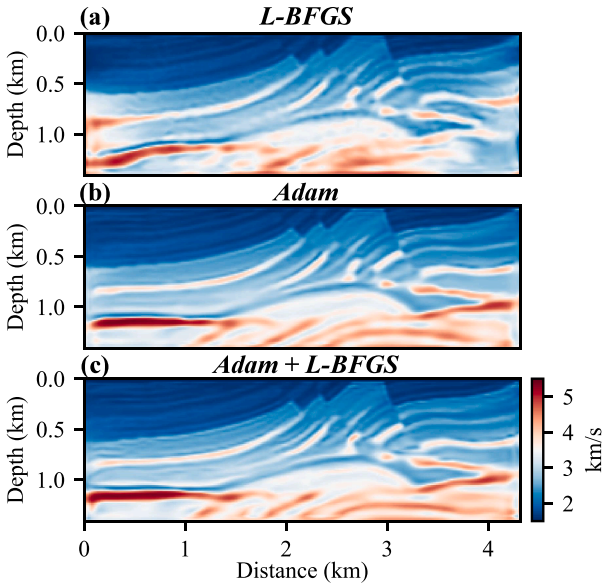


Fig. 21. Comparison of inversion results using different optimizers. (a) The result using L-BFGS. (b) The result using Adam. (c) The result using Adam + L-BFGS.

Limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) is considered as an effective second-order algorithm suitable for large-scale optimization problems (Nocedal & Wright, 2006). By approximating the inverse Hessian matrix, L-BFGS leverages second-order curvature information for more efficient parameter updates. For RFWI, the optimization process using L-BFGS can be reformulated as:

$$\boldsymbol{\theta}^{t+1} = \boldsymbol{\theta}^t - \eta(\mathbf{B}^t)^{-1}\nabla\mathcal{L}(\boldsymbol{\theta}^t) \tag{19}$$

where $(\mathbf{B}^t)^{-1} \approx [\mathbf{H}(\theta)^t]^{-1} = \mathbf{V}\Lambda^{-1}\mathbf{V}^T$ denotes the approximated inverse Hessian matrix. Similarly to the analysis in Section 4.1.1, we express the network parameter update vector as a weighted sum of the eigenvectors. It is worth noting that we retain the term $(\mathbf{B}^t)^{-1}$ in the following formulation to emphasize the contribution of L-BFGS in improving the ***convergence*** of RFWI:

$$\Delta\boldsymbol{\theta}^t = \theta^{t+1} - \theta^t = (\mathbf{B}^t)^{-1}\sum_{i=1}^{M}\alpha_i^t\mathbf{v}_i^t \tag{20}$$

$$\alpha_i^t = \frac{1}{\lambda_i^t}(\mathbf{v}_i^t)^T\Delta\boldsymbol{\theta}^t \tag{21}$$

Clearly, the $(\mathbf{B}^t)^{-1}$ approximated in L-BFGS can increase the step size along the eigenvectors associated with small eigenvalues, which typically correspond to high-frequency information, as demonstrated in Section 4.2. In other words, L-BFGS can enhance the sensitivity of RFWI to high-frequency information. This is consistent with our thinking to improve RFWI ***convergence***. For RFWI, adequately fitting low-frequency information in the early stages is essential to alleviate susceptibility to local minima. However, using L-BFGS alone in RFWI may undermine this crucial optimization characteristic. Therefore, we propose a hybrid optimizer that applies Adam followed by L-BFGS (referred to as Adam + L-BFGS). This technique allows Adam to establish a robust low-frequency velocity model, followed by L-BFGS to enhance the ***convergence*** of high-frequency components.

The inversion results and corresponding loss curves obtained for different optimizers are presented in Figs. 21 and 22, respectively. For Adam and L-BFGS optimizer, 4000 iterations are performed, whereas for the Adam + L-BFGS hybrid optimizer, only 1100 iterations are conducted, with 600 for Adam and 500 for L-BFGS. From the inversion results, using L-BFGS alone leads to pronounced structural artifacts in the model, while Adam and Adam + L-BFGS achieve satisfactory reconstructions. As shown in Fig. 22, employing L-BFGS alone leads to a rapid reduction in data loss, yet the model loss exhibits the poorest performance. This indicates that, in the absence of an initial model, using L-BFGS alone causes the optimization to become trapped in a local minimum. The model loss curve of the optimization process using Adam + L-BFGS exhibits a faster decline after the 600-th epoch when the optimizer switches to L-BFGS. To enhance the reliability of the quantitative results, we repeat all experiments five times using different random seeds and report all evaluation metrics in the form of mean ± standard deviation. The statistics comparison in Table 5 shows that Adam + L-BFGS achieves inversion results comparable to Adam with the least computational time. Overall, both qualitative and quantitative results demonstrate that the Adam + L-BFGS hybrid optimizer further improves RFWI ***convergence*** while maintaining robustness in the ideal acquisition scenario.

5.2. Robustness improvement

5.2.1. Connection between Hessian trace and optimization characteristics

According to the frequency principle of DNNs and our experimental results in Section 3.3.2, noise overfitting in the late stage of inversion is considered as the primary factor limiting the RFWI performance. In the field of imaging inverse problems, two effective methods are employed to mitigate noise overfitting:

(1): Ulyanov et al. (2018) suggested monitoring the discrepancy between the network output and the ground truth, and terminating the training when the output reaches its optimal performance. This method is referred to as 'early stopping'.

(2): Heckel et al. (2019) proposed an under-parameterized network architecture (i.e., the number of network parameters is smaller than the output dimension) to construct concise image representations. They limited the network's ability to represent complex noise by decreasing the number of network parameters.

However, these methods remain limited in the context of RFWI. For the first method, the primary limitation is that the ground truth (i.e., the true velocity model) is typically unavailable in practical applications. For the second method, subsurface models exhibit much greater structural complexity than natural images, making it difficult for an under-parameterized network to accurately reconstruct intricate geological structures, as demonstrated in Wu and Ma (2025). Building on these considerations, we suggest that a feasible method is to reduce the high-frequency sensitivity (or convergence rate) of over-parameterized DNNs, thereby achieving a desirable balance between representing complex structures and mitigating noise overfitting.

As mentioned in Section 2.3, there is a connection between the sharpness of the optimization trajectory and the generalization of DNNs in classification tasks. Moreover, the three optimizers discussed above (L-BFGS, Adam, and Adam + L-BFGS) drive the optimization along distinct trajectories, resulting in pronounced differences in sensitivity to high-

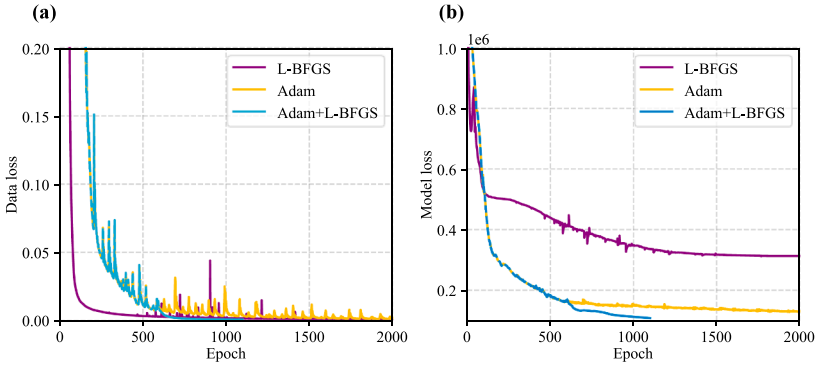


Fig. 22. Loss curves of inversion processes using different optimizers. (a) The process using L-BFGS. (b) The process using Adam. (c) The process using Adam + L-BFGS.

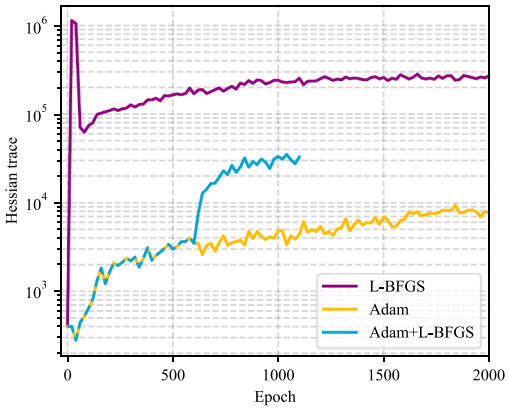


Fig. 23. The evolution of the Hessian trace during the inversion process with different optimizers.

frequency information. This inspires us to investigate the connection between the sharpness of the optimization trajectory and the optimization characteristics in the context of the RFWI framework based on these three optimizers.

Following previous studies (Sankar et al., 2021; Yao et al., 2020), we adopt the Hessian trace as an indicator of the sharpness of the optimization trajectory. Fig. 23 shows the evolution of the Hessian trace using three different optimizers. It can be observed that, compared with L-BFGS, Adam drives the optimization along a flatter trajectory, reflected by the smaller Hessian trace. Moreover, our comparative results in Section 5.1 clearly demonstrate that the optimization process using Adam exhibits a slower convergence toward high-frequency information than using L-BFGS. Therefore, we empirically conclude that a flatter optimization trajectory results in more robust optimization characteristics, enabling adequate fitting of low-frequency information while suppressing sensitivity to high-frequency components.

5.2.2. Hessian trace-based regularization

Motivated by this empirical conclusion, we propose a Hessian trace-based regularization technique that guides a flatter optimization trajectory to further improve the RFWI ***robustness***. Specifically, we explicitly penalize the sharpness of optimization trajectory by incorporating the Hessian trace as a regularization term into the loss function, which yields the modified formulation:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\left\|\mathbf{F}(\mathcal{N}(\mathbf{z},\boldsymbol{\theta})) - \mathbf{d}_{\text{obs}}\right\|_2^2 + \gamma \cdot Tr[\mathbf{H}(\theta)] \tag{22}$$

where γ denotes the regularization coefficient, and $Tr[\mathbf{H}(\theta)]$ [hereafter abbreviated as $Tr(\mathbf{H})$] represents the trace of the Hessian matrix $\mathbf{H}(\theta)$. The Hessian trace can be estimated using Hutchinson's method (Avron & Toledo, 2011) without explicitly constructing the Hessian matrix. Consider a random vector $\mathbf{x}$ whose entries are independently and identically distributed (i.i.d.) according to a Rademacher distribution. Then, the Hessian trace satisfies:

$$Tr(\mathbf{H}) = Tr(\mathbf{HI}) = Tr(\mathbf{H}\mathbb{E}[\mathbf{x}\mathbf{x}^T]) = \mathbb{E}[Tr(\mathbf{H}\mathbf{x}\mathbf{x}^T)] = \mathbb{E}[\mathbf{x}^T\mathbf{H}\mathbf{x}], \tag{23}$$

where $\mathbf{I}$ denotes the identity matrix with the same dimensions as $\mathbf{H}(\boldsymbol{\theta})$, and $\mathbb{E}[\cdot]$ denotes the expectation operator. Based on the this formulation, the Hessian trace can be estimated using Algorithm 2. By retaining the computational graph, the gradient of the Hessian trace with respect to the network parameters can be computed via backpropagation.

Algorithm 2: Hutchinson's method for estimating Hessian trace

Input : Network parameters θ; Loss function $\mathcal{L}$; Number of samples N;
Output: Estimated Hessian trace $Tr(\mathbf{H})$;

Compute gradient via backpropagation: $\nabla\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\theta}$;
for $i = 1$ **to** N **do**
  Draw a random vector $\mathbf{x_i}$ from Rademacher distribution;
  Compute: $\nabla\mathcal{L}^T\mathbf{x}_i$;
  Compute: $\mathbf{H}\mathbf{x}_i = \partial(\nabla\mathcal{L}^T\mathbf{x}_i)/\partial\boldsymbol{\theta}$;
  Compute and record: $\mathbf{x}_i^T\mathbf{H}\mathbf{x}_i$
$Tr(\mathbf{H}) \leftarrow \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_i^T\mathbf{H}\mathbf{x}_i$;
return Tr($\mathbf{H}$);

We refer to this regularization technique as Hessian trace regularization (HTR) and the RFWI method with HTR as HTR-RFWI. We still choose Adam as the optimizer. The number of samples is set to 5, initial learing rate is set to 2e-3, and the regularization coefficient γ is set to 1e-4. Considering the computationally intensive nature of Hessian trace calculations, we only employ the HTR technique at periodic intervals of 10 iterations. To validate the effectiveness of the proposed HTR technique, we compare HTR with other regularization schemes, including total variation (TV) regularization, Tikhonov regularization, and adaptive sharpness-aware minimization (ASAM) regularization (Kwon et al., 2021). ASAM searches for the local worst-case perturbation (i.e., the perturbation that maximizes the loss) along the gradient direction in the loss landscape and updates parameters using the perturbed gradients, guiding the optimization trajectory toward low-curvature regions. For the hyperparameter selection, the regularization coefficients for TV-RFWI and Tikhonov-RFWI are set to 3e-5 and 4e-8, respectively. For ASAM-RFWI, the learning rate is set to 2e-3 and the perturbation radius is set to 0.02, respectively.

We consider using degraded data to evaluate these methods under challenging scenarios. For simulation of degraded data, the Ricker wavelet with a dominant frequency of 12 Hz serves as the source. In addition, we remove low-frequency components below 4 Hz and merge Gaussian white noise with mean 0 and standard deviation of $2\sigma_0$. In this way, the availability of low-frequency data is reduced, and the high-

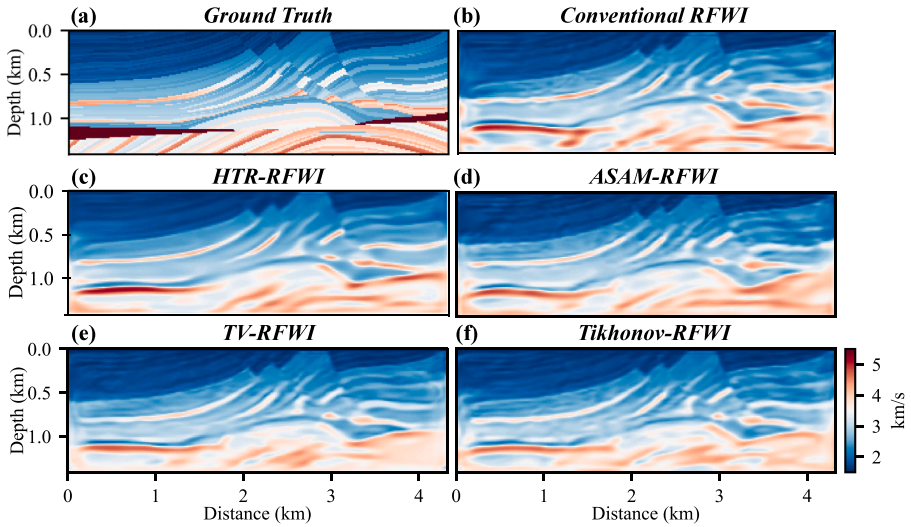


Fig. 24. Comparison of optimal RFWI inversion results with different regularization schemes. (a) Ground truth. (b) Conventional RFWI. (c) HTR-RFWI. (d) ASAM-RFWI. (e) TV-RFWI. (f) Tikhonov-RFWI.

frequency components become dominated by noise. All experiments are repeated five times using different random seeds, and each experiment involves 2600 optimization iterations.

To display the upper-bound performance that each method can achieve under controlled experimental conditions, we select the optimal inversion result of each implementation for comparison. Since RFWI may gradually overfit noise in the later optimization stages, the optimal inversion performance is typically achieved before the final optimization iteration. Fig. 24 and Table 6 present the qualitative and quantitative comparisons of the optimal RFWI inversion results with different regularization schemes, respectively. As can be observed, the inversion result of conventional RFWI contains significant artifacts, and shows limited reconstruction of the high-frequency interface features at the bottom of the model. Tikhonov-RFWI achieve smoother models with lower MSE than conventional RFWI, but at the cost of suppressing high-frequency details, leading to lower SSIM. TV-RFWI improves the clarity of geological interfaces to some extent, but provides limited improvement in the overall smoothness of the model. In contrast, sharpness-based regularization schemes, including ASAM-RFWI and HTR-RFWI, improve both global smoothness and high-frequency fidelity. Among them, HTR-RFWI outperforms ASAM-RFWI. This difference can be attributed to their optimization mechanisms: ASAM evaluates gradients at perturbed parameters, which may reduce consistency with the underlying physics constraints and introduces sensitivity to the neighborhood-size hyperparameter. In contrast, HTR relies on gradients computed at the current parameter state via backpropagation, leading to more stable and physically consistent updates. It is worth noting that HTR-RFWI and ASAM-RFWI require longer computational times than other methods. For HTR-RFWI, the increased computational cost mainly results from the need to retain the computational graph and perform backward propagations to compute the Hessian trace and its gradient with respect to the network parameters. For ASAM-RFWI, performing an additional forward and backward propagation at the perturbed parameter point increases the computational time. Nevertheless, HTR-RFWI achieves the best performance in terms of MSE, SSIM, and SNR with an acceptable increase in computational time.

The evolution curves of the Hessian trace and the model loss show the refinement of the HTR technique for the inversion ***robustness***. As shown in Fig. 25(a), the HTR technique significantly reduces the Hessian trace of the inversion process, that is, it guides the inversion to occur on a flatter trajectory. Fig. 25(b) shows that HTR-RFWI method exhibits a more stable optimization process. Overall, with appropriate hyperparameter settings, HTR technique enables RFWI to mitigate noise overfitting while still effectively representing high-frequency components.

To avoid introducing oracle and better reflect practical scenarios, all RFWI methods are performed with a fixed number of 1500 iterations. The statistical comparison is shown in Table 7. It can be observed that each regularization scheme improves the performance of RFWI to different degrees. The proposed HTR-RFWI achieves the best performance among all compared methods in terms of all evaluation metrics. This result further demonstrates the effectiveness of the HTR scheme.

5.3. Generalization verification

To further evaluate the generalization of the proposed methods, we extend our evaluation to the widely used Overthrust benchmark model, as shown in Fig. 26(a). Except for the benchmark, all experimental settings are kept identical to those used in the corresponding experiments presented above.

Convergence improvement. Fig. 26 and Table 8 respectively present the qualitative and quantitative comparisons of the inversion results obtained using different optimizers on the Overthrust model. Taken together, these results demonstrate that the Adam–L-BFGS optimizer can achieve inversion results comparable to those obtained by Adam with reduced computational cost, while yielding superior inversion performance compared with using L-BFGS alone.

Table 6
Statistics comparison of optimal RFWI inversion results with different regularization schemes.

	MSE ↓ ($\times 10^6$)	SSIM↑	SNR↑	Computational Time (min)↓
Conventional RFWI	0.1923±0.0067	0.6185±0.0071	16.7457±0.1512	**7.83**
HTR-RFWI	**0.1456±0.0087**	**0.6425±0.0044**	**17.9671±0.2544**	15.19
ASAM-RFWI	0.1708±0.0092	0.6259±0.0185	17.2620±0.2356	15.62
TV-RFWI	0.1869±0.0072	0.6193±0.0073	16.8671±0.1640	7.89
Tikhonov-RFWI	0.1776±0.0044	0.6167±0.0162	17.0854±0.1070	7.88

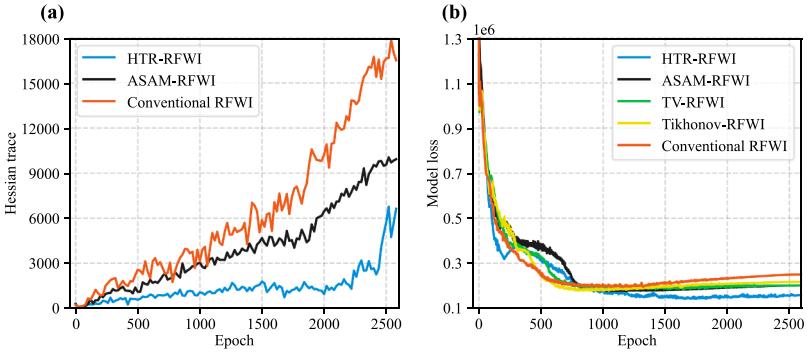


Fig. 25. Comparison of the Hessian trace and model loss evolution of RFWI methods with different regularization schemes. (a) Hessian trace. (b) Model loss.

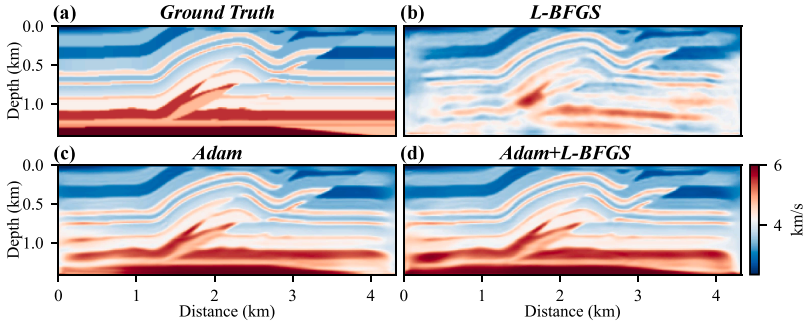


Fig. 26. Comparison of inversion results using different optimizers on the Overthrust model. (a) Ground truth. (b) The result using L-BFGS. (c) The result using Adam. (d) The result using Adam + L-BFGS.

Table 7
Statistical comparison of RFWI inversion results with different regularization schemes after 1500 iterations.

	MSE ↓ ($\times 10^6$)	SSIM ↑	SNR ↑
Conventional RFWI	0.2236±0.0120	0.6040±0.0176	16.2794±0.2598
HTR-RFWI	**0.1585±0.0177**	**0.6279±0.0168**	**17.5407±0.3431**
ASAM-RFWI	0.1790±0.0101	0.6186±0.0177	17.0360±0.2762
TV-RFWI	0.1929±0.0068	0.6123±0.0059	16.6996±0.1488
Tikhonov-RFWI	0.1872±0.0103	0.6092±0.0165	16.9411±0.1548

Table 8
Statistics comparison of inversion results using different optimizers on the Overthrust model.

	MSE ↓ ($\times 10^6$)	SSIM↑	SNR↑	Computational Time (min)↓
L-BFGS	0.3832	0.6349	16.1988	63.77
Adam	0.0448	**0.8834**	22.7411	12.00
Adam + L-BFGS	**0.0307**	0.8749	**23.4283**	**10.18**

Robustness improvement. We further evaluate the generalization of HTR under the challenging scenario, and the qualitative and quantitative comparisons between conventional RFWI and HTR-RFWI are illustrated in Fig. 27 and Table 9. Compared with conventional RFWI, the inversion results obtained by HTR-RFWI exhibit fewer artifacts and reveal clearer and more accurate subsurface interfaces.

Table 9
Statistics comparison of inversion results between conventional RFWI and HTR-RFWI.

	MSE ↓ ($\times 10^6$)	SSIM ↑	SNR ↑
Conventional RFWI	0.1247	0.7155	20.1183
HTR-RFWI	**0.0778**	**0.7622**	**23.6685**

Overall, the results obtained on the Overthrust model are highly consistent with those observed on the Marmousi model, providing further evidence for the generalization of the proposed methods and supporting the main conclusions presented in this study.

6. Discussion

6.1. Influence of configuration components on inversion results

As claimed by many previous studies, the choice of configuration components appears to affect the inversion results. In this section, we analyze the influence of different configuration components, including network input, initialization strategy and network architecture, on the inversion results.

6.1.1. Influence of network input

For network input, the most common choice is the random tensor sampled from a standard Gaussian distribution. However, some re-

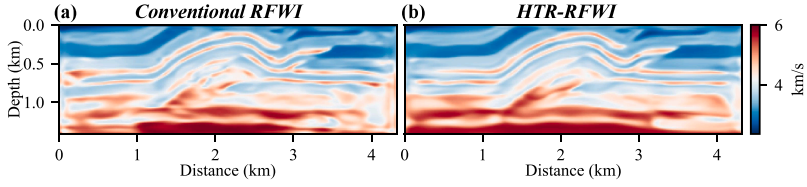


Fig. 27. Comparison of inversion results between conventional RFWI and HTR-RFWI method on the Overthrust model. (a) Conventional RFWI. (b) HTR-RFWI.

searchers suggested that using shot gathers as input can introduce geological prior information, thus improving inversion results (Dhara & Sen, 2022; Jiang et al., 2023). We aim to understand the influence of network input on inversion results from the perspective of the loss landscape.

Since the network input constitutes one of the fundamental factors determining the geometry of loss landscape, it is essential to align optimization processes with different network input to an identical loss landscape. In our analysis, both the network architecture and the weights initialization are kept identical in all cases. We assume that a virtual input and a virtual convolutional layer exist before the actual input, so the actual input can be regarded as feature maps of the virtual convolutional layer in the feed-forward process. When all inversion processes adopt same virtual input, they become unified within the same loss landscape. The differences in actual input can be fully attributed to differences in the weights of the virtual convolutional layer. Therefore, the network input is essentially a random initialization way of the network weights, which defines different initial state for optimization. Theoretical analyses of DNNs indicate that, under well-chosen hyperparameter settings, the performance of local minima closely approximates that of the global minimum in the loss landscape (Kawaguchi, 2016). Therefore, we conclude that network input does not play a determining role in inversion performance.

6.1.2. Influence of initialization strategy

Likewise, the pretraining strategy via supervised learning can be regarded as a special initialization. It is intuitively clear that pretraining will define a beginning that is closer to the nearby convergent point, and thus achieve faster convergence. Although our experimental results presented in Fig. 6 support this statement, the inversion results obtained with the pretraining strategy unexpectedly appear to be of lower quality in our framework. We think that differences in the properties of the gradients can explain the convergence to suboptimal local minimum observed when employing the pretraining strategy. In the pretraining stage, the gradients calculated to update the network weights are obtained from the misfit of the pixel values (i.e., velocity values) between the inversion result and the initial model. In contrast to this flexible and accurate gradient, the computation of the gradient in the formal inversion process is based on complicated physical relations. This difference likely defines an initial state that deviates from a physically accurate solution in the pretraining stage. However, this phenomenon is not absolute and may be closely related to the network architecture.

6.1.3. Influence of network architecture

Due to the combination of spatial and semantic information through skip connection, U-Net demonstrates enhanced performance in reconstructing detailed features compared to AE architecture in computer vision applications. However, the results in Section 3.3.1 demonstrate that the AE architecture achieves better inversion performance compared to U-Net within the RFWI framework. To illustrate this phenomenon, in the U-Net architecture, we extract the average sum of the feature maps from the convolutional layers numbered 1 and 2 in Fig. 2, as shown in Fig. 28. As we can see, the feature maps located in the shallow layers maintain similar spatial features as the network input, so the corresponding deep feature maps similarly exhibit these patterns. Such geologically unreasonable features will interfere with inversion results, especially in challenging scenarios. Consequently, the smooth inversion results generated by the AE architecture appear to be more compatible with the requirements of RFWI, particularly in scenarios demanding enhanced robustness.

6.2. Limitations and future work

Building upon the findings of this study, we identify that several important challenges remain unresolved and deserve further investigation.

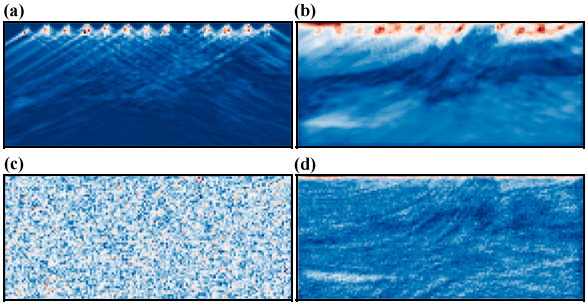


Fig. 28. Comparison of average sum of the feature maps of the convolutional layers numbered 1 and 2 in Fig. 2. The feature maps presented are all extracted from the unet network. (a) and (b) Layer 1 and Layer 2 with the data as input. (c) and (d) Layer 1 and Layer 2 with the Gaussian noise as input.

Although the proposed HTR technique demonstrates promising performance, it should not be regarded as the optimal regularization scheme for RFWI method. Developing more effective and efficient regularization schemes tailored to the optimization characteristics of RFWI requires a deeper understanding of its underlying loss landscape. Furthermore, this study focuses exclusively on the 2D acoustic setting, which may differ substantially from elastic-wave and 3D inversion scenarios. And this study primarily focuses on the DR-RFWI paradigm. Because the role of the DNN during the inversion process in INR-RFWI differs from that in DR-RFWI, the proposed strategy may not necessarily be applicable to INR-RFWI. Consequently, the conclusions and methods presented in this work may require further validation and potential adaptation before being applied to these settings. Extending the proposed methods to INR-RFWI paradigm and these more complex inversion scenarios will be a major focus of our future research.

7. Conclusion

In this work, we present insights into two key questions in a Hessian perspective. Regarding the first question, We employ the *Hessian spectrum* to characterize the optimization dynamics and analyze its physical interpretation. And the combined insights from the two perspectives provide a profound understanding of the optimization characteristics of RFWI. We demonstrate that the constraints imposed by DNN drive the RFWI optimization toward stable eigenvectors linked to large eigenvalues, thereby suppressing the sensitivity of the inversion process to high-frequency components.

In response to the second question, based on the discovered optimization characteristics of RFWI, we propose two generalizable techniques to further improve ***convergence*** and ***robustness***, respectively. To improve ***convergence***, we propose the Adam + L-BFGS hybrid optimizer, where Adam is used to establish a stable low-frequency model, and L-BFGS accelerates the fitting of high-frequency information by increasing step sizes along eigenvectors corresponding to small eigenvalues. For ***robustness*** improvement, we empirically demonstrate that the optimization process along the flatter trajectory exhibits reduced sensitivity to high-frequency information, which motivates the proposal of the Hessian trace-based regularization technique.

In summary, our work further investigates the RFWI framework through the novel Hessian perspective. We believe that these findings can deepen the community's understanding of the RFWI framework and inspire future research endeavors.

CRediT authorship contribution statement

Peng Jiang: Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization; **Sihan Chen:** Writing – original draft, Validation, Methodology, Investigation, Conceptualization; **Yuxiao Ren:** Writing – review & editing, Validation, Su-

pervision, Data curation; **Senlin Yang:** Validation, Supervision, Formal analysis; **Bin Liu:** Visualization, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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